

Product Reference Manual (V1.0)

Description

The **UNIT Capacitive Touch Sensor** replaces traditional buttons and mechanical switches by converting a simple human touch into a precise digital signal. Powered by the **TTP223B** capacitive sensing IC, this module continuously monitors its flat electrode pad and outputs a clear **HIGH logic signal** the instant a touch is detected.

Designed for **reliable, debounce-free touch detection**, it performs consistently across a variety of applications — from **sleek control panels** and **wearable interfaces** to **touch-activated lighting systems**. With **minimal wiring requirements** and **ultra-low standby current**, the UNIT Capacitive Touch Sensor offers an elegant and energy-efficient alternative to mechanical input systems.

DevLab format compatibility.

Simplicity and compatibility are the core principles of the **DevLab form factor**. This standard defines a **compact and communication-optimized board layout**, ensuring straightforward connection and interoperability among DevLab modules.

By adhering to this format, the UNIT Capacitive Touch Sensor guarantees **efficient prototyping, ease of integration**, and **unified accessibility** across a wide ecosystem of devices and development platforms.



Key Features

- TTP223B capacitive sensing IC for precise touch detection.
- Auto-calibration and noise filtering for stable operation.
- Digital HIGH output when touch is detected.
- Selectable modes: Momentary or Toggle via solder jumper.
- JST 1.0 mm connector for easy integration.

Hardware Features

- Operating Voltage: 2.0 V – 5.5 V
- Typical Supply Current:
- Low Power Mode: 1.5 μ A (typ), 3.0 μ A (max)
- Fast Mode: 3.5 μ A (typ), 7.0 μ A (max)
- Response Time (@ 3 V): 60 ms (Fast), 220 ms (Low Power)
- Output Logic: Digital CMOS output (active HIGH by default)
- Operating Modes: Momentary / Toggle (selectable via jumper)
- Touch Pad Type: Large silver electrode, single channel
- Indicator: Orange LED (touch status)
- Interface Connector: JST 1.0 mm 4-pin (VCC, GND, OUT, SHIELD)
- Form Factor: DevLab-compatible compact module
- Operating Temperature: -40°C to $+85^{\circ}\text{C}$
-

Software Support

- **PlatformIO / VS Code:** Offers a professional development

environment with **multi-board build automation**, advanced debugging, and version control integration for streamlined workflows.

- **MicroPython**

Applications

- **Touch-activated switches and control panels**
- **Smart home interfaces** (lighting, appliances, automation)
- **Wearable devices and portable electronics**
- **Interactive art and installation projects**
- **DIY electronics and education kits**
- **Embedded systems** requiring compact, button-free input

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1 The Board

1.1 Accessories

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2 Ratings

2.1 Recommended Operating Conditions

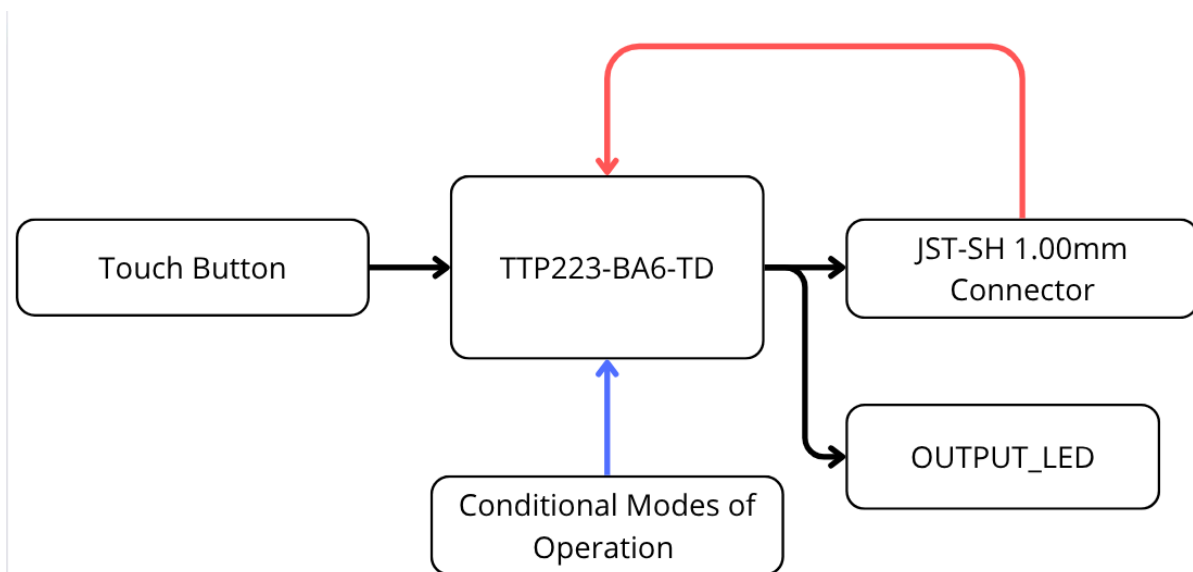
Symbol	Description	Min	Typ	Max	Unit
V_{DD}	Operating voltage	2.0	3.0	5.5	V
I_{DD-LP}	Supply current, Low Power mode	—	1.5	3.0	μ A
I_{DD-F}	Supply current, Fast mode	—	3.5	7.0	μ A
t_{RESP}	Response time (@3 V)	—	60	220	ms
C_{EXT}	External capacitance range (sensitivity)	0	—	50	pF
V_{OH}	Output HIGH voltage	$0.7 \times V_{DD}$	—	VDD	V
V_{OL}	Output LOW voltage	0	—	$0.3 \times V_{DD}$	V
T_{OP}	Operating temperature range	−40	—	+85	°C

3 Functional Overview

The UNIT Capacitive Touch Sensor detects minor changes in capacitance that occur when a finger makes contact with its conductive pad. These variations are managed by the TTP223B controller, which provides a clean and stable digital output. The circuit automatically adjusts itself to maintain accuracy under different environmental conditions, such as shifts in humidity or temperature. When the measured capacitance passes a defined threshold, the OUT-pin switches to a HIGH logic level, indicating that a valid touch has been registered.

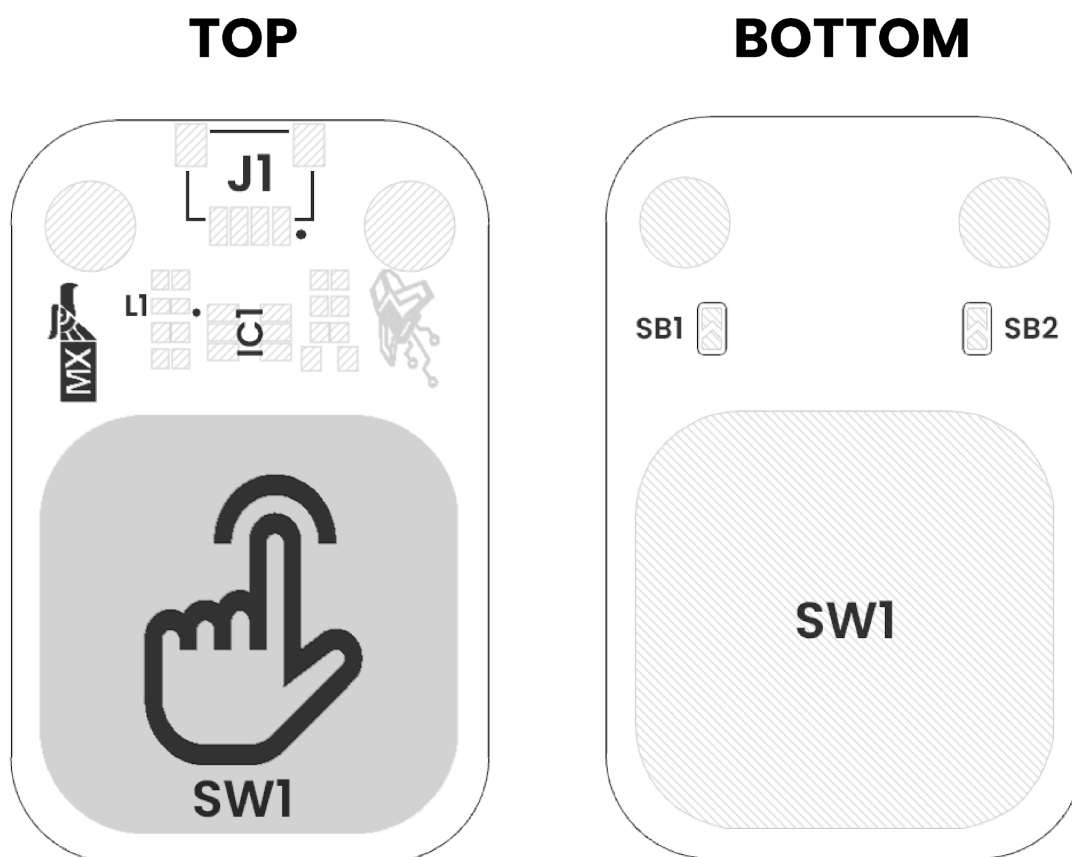
The module can operate in Momentary Mode, where the output remains HIGH only during touch, or in Toggle Mode, where the state alternates between HIGH and LOW with each press. Users can fine-tune response speed, output polarity, and power behavior using the configuration pads. Designed to be compact and energy-efficient, the sensor integrates a status LED and serves as a reliable, long-life replacement for mechanical switches in embedded electronics and interactive control applications.

3.1 Block Diagram



Block Diagram of Capacitive Touch Sensor

3.2 Board Topology



Views of Board Topology

Views of Capacitive Touch Sensor Topology

Table 3.2.1 - Components Overview

Ref.	Description
SW	Capacitive Touch Button
L1	Built-In LED
IC1	TTP223-BA6-TD Touch Detector
J1	QWIIC Connector (JST 1 mm pitch) for I2C
SB1	Solder Bridge for Mode Selection
SB2	Solder Bridge for Logic Level Selector

3.3 UNIT Capacitive Touch Sensor

The **UNIT Capacitive Touch Sensor** is a digital input module based on the **TTP223** touch pad detector IC, designed to replace traditional mechanical buttons with a solid-state capacitive surface. It supports one touch key with high stability and low power consumption, operating from **2.0 V to 5.5 V**.

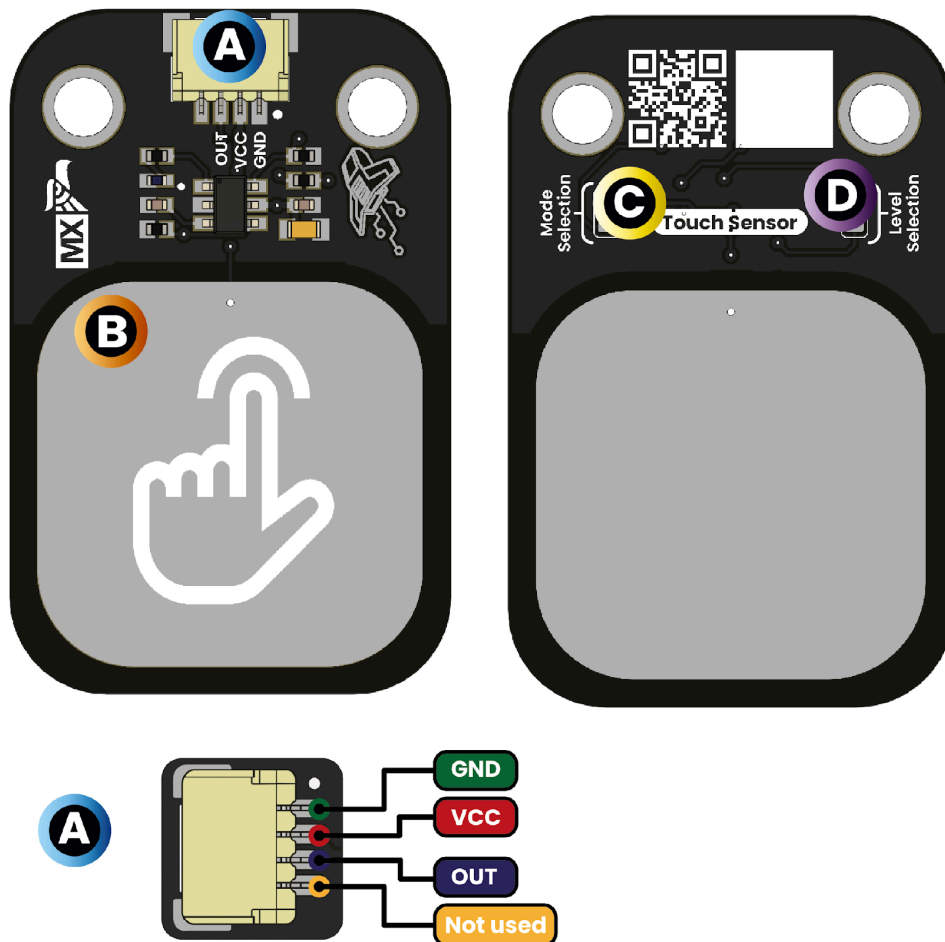
The onboard **TTP223B** controller continuously senses changes in capacitance on its electrode pad, providing a clean **digital output (HIGH or LOW)** through both **CMOS** and **open-drain** modes. Functional behavior such as **momentary or toggle mode**, **active-high or active-low output**, and **power-saving options** can be selected via solder jumpers.

With a fast response time (≈ 60 ms) and auto-calibration every 4 seconds, the UNIT sensor ensures reliable touch detection for **user interfaces**, **smart devices**, and **IoT applications**.

4 Connectors & Pinouts

4.1 General Pinout

PINOUT



Description:

- | | |
|--|---|
|  Supply voltage |  Touch Pad |
|  GND |  Mode selection |
|  Output |  Level selection |

Capacitive Touch Sensor Module *General Pinout*

4.2 Pinout General Description

Pin Label	Description
VCC	Power supply (3.3V or 5V)
GND	Ground
PIN	Out-pin
-	-

5 Board Operation

5.1 Getting Started with Micropython

The **DevLab Capacitive Touch Sensor** simplifies user interaction by converting a simple touch into a precise digital signal — no buttons, no moving parts. Powered by the **TTP223B capacitive sensing chip**, it continuously monitors its flat electrode pad and reports a “touch detected” event through a clean **HIGH logic output**. You can easily customize its behavior by soldering configuration pads to select between **momentary** and **toggle** modes, or invert to the output level.

What You'll Need

- DevLab **TTP223B Capacitive Touch Sensor**
- Microcontroller board (e.g., **ESP32** or **RP2040**)
- JST 4-Pin Pitch 1.00mm
- USB cable
- MicroPython firmware and [Thonny IDE](#)



Mode & Level Selection

Mode Selection (TOG)	Level Selection (AHLB)	Output Type	Behavior
0	0	Momentary, Active HIGH (Factory Default)	Output goes HIGH while touched
0	1	Momentary, Active LOW	Output goes LOW while touched
1	0	Toggle, Active HIGH	Touch toggles output ON/OFF
1	1	Toggle, Active LOW	Touch toggles inverted (starts HIGH)

- **Mode Selection (TOG):**
 - o 0 → Momentary (active only while touched)
 - o 1 → Toggle (changes state each touch)
- **Level Selection (AHLB):**
 - o 0 → Active HIGH
 - o 1 → Active LOW (inverted logic)

Bridge each solder pad pair with a small drop of solder to set the desired mode.

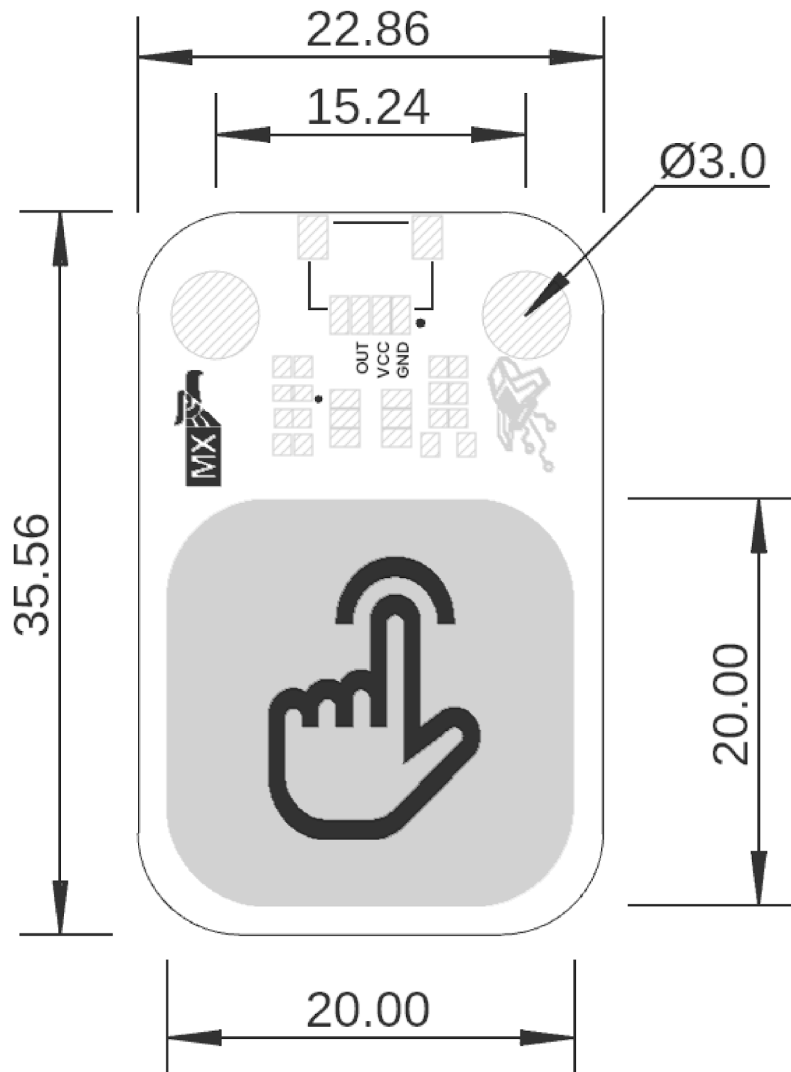
MicroPython Example

The official example is available in the DevLab repository:

[sensor_touch.py — UNIT-Electronics-MX/unit_devlab_capacitive_touch_sensor](https://github.com/UNIT-Electronics-MX/unit_devlab_capacitive_touch_sensor/blob/main/sensor_touch.py)

```
File Edit View Run Tools Help
<untitled> * x
1 from machine import Pin
2 import time
3
4 # Using GPIO12
5 touch_pin = Pin(12, Pin.IN, Pin.PULL_DOWN)
6
7 pulse_count = 0
8
9 def on_pulse(pin):
10     global pulse_count
11     pulse_count += 1
12     print("Pulse detected! Total:", pulse_count)
13
14 # Interrupt on rising edge (adjust if your sensor is active-low)
15 touch_pin.irq(trigger=Pin.IRQ_RISING, handler=on_pulse)
16
17 print("Waiting for pulses from capacitive sensor on GPIO12...")
18
19 # Main loop
20 while True:
21     time.sleep(1) # Avoid using CPU, everything handled by interrupt
22
```

6 Mechanical Information



Mechanical dimensions in millimeters (mm)

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	Link
Documentation	https://github.com/UNIT-Electronics-MX/unit_devlab_capacitive_touch_sensor
Wiki	https://wiki.uelectronics.com/wiki/devlab-capacitive-touch-sensor
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download
TTP223 - DataSheet	https://uelectronics.com/wp-content/uploads/2018/02/AR0326-TTP223B-Boton-Tactil-Capacitivo-Datasheet.pdf

9 Appendix

9.1 Schematic (https://github.com/UNIT-Electronics-MX/unit_devlab_capacitive_touch_sensor/blob/main/hardware/unit_sch_V_0_0_1_ue0099_Sensor_Touch.pdf)

